

SHIVAM PATIL

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SUMMARY

Software Engineer with 1.7 years building production Python/FastAPI services and RAG retrieval systems at TCS, and low-latency market-data, order-entry, and distributed systems in Rust, C++, and Go through self-directed projects.

PROFESSIONAL EXPERIENCE

Software Development Engineer — Tata Consultancy Services

Jan 2025 – Present

- Designed and shipped 13+ REST APIs (Python/FastAPI) for a production system serving 10,000+ requests/day, cutting response latency by 33%.
- Optimized SQL queries and data-processing pipelines, improving application performance by 29% while holding reliability at scale.
- Raised RAG answer accuracy by 19% on a production RBI chatbot by tuning vector-search relevance and building the automated evaluation framework that measured it.
- Implemented authentication and authorization against enterprise access-control standards; collaborated on code reviews upholding testability and maintainability.

PROJECTS

Low-Latency Market Data & Order Entry Stack

Stack: Rust, C++, FIX 4.4, UDP multicast (redundant A/B), SBE-style binary encoding, Docker

- Built an end-to-end trading stack: a price-time-priority matching engine publishing a binary market-data feed over redundant A/B UDP multicast channels, with configurable packet-loss injection, a 2-second snapshot cycle, and a TCP replay service for recovery.
- Engineered a Rust feed handler performing A/B feed arbitration, sequence-gap detection, and snapshot-based recovery into MBP/MBO order books — sustaining 1M+ messages/sec at ~100ns decode and ~200ns book update, with zero heap allocations per message verified by a counting allocator.
- Implemented a FIX 4.4 order gateway with a full session layer (logon, heartbeats, resend/gap-fill, durable sequence persistence) reconciling order state across a hard process restart, plus a risk service enforcing pre-trade limits on an allocation-free path.

Distributed Rate Limiter & API Gateway

Stack: Go, gRPC/REST, Redis, PostgreSQL, Docker, Azure Kubernetes Service, Prometheus/Grafana, React

- Engineered a distributed rate-limiting gateway in Go using token-bucket and sliding-window algorithms, sharded via consistent hashing with Raft-based leader election for automatic failover.
- Exposed gRPC/REST APIs over Redis and PostgreSQL; load-tested to 45,000 req/sec at sub-8ms p99 with k6, deployed on AKS with Prometheus/Grafana and a React dashboard for live traffic and quota state.

High-Performance LLM Inference Server

Stack: Rust, Python (PyO3 bindings), gRPC, Docker, React

- Built an LLM inference server in Rust implementing continuous batching and KV-cache management to maximize concurrent request throughput, exposed over a gRPC interface.
- Added dynamic batching and quantized (INT8/FP16) model loading — throughput up 230% and p99 latency down 42% versus naive single-request serving — with a React dashboard for token throughput, batch utilization, and memory footprint.

TECHNICAL SKILLS

Proven in production at work — *shipped and operated at Tata Consultancy Services, Jan 2025 – Present*

Python, FastAPI, REST: 13+ APIs on a production system serving 10,000+ requests/day

SQL, data pipelines, authN/authZ: query and pipeline optimization under production load; enterprise access-control standards

RAG, vector search, evaluation: relevance tuning plus the automated eval harness that measured it

Proven outside work — *designed, built, and benchmarked end-to-end in the projects above*

Rust, C++: allocation-free Rust feed handler at 1M+ msg/sec, matching engine, FIX 4.4 session layer, pre-trade risk checks

Go, gRPC, Redis, PostgreSQL: consistent hashing, Raft leader election, k6 load testing at 45,000 req/sec

Docker, Kubernetes (AKS), Prometheus/Grafana: containerized deploys, metrics, and operator dashboards

TypeScript, React, Next.js: shivamsfolio.com — the site linked above, built and deployed solo

Concurrency & systems: lock-free and zero-allocation hot paths, binary wire protocols, distributed consensus

EDUCATION & CERTIFICATIONS

- B.Tech, Artificial Intelligence and Data Science — Pune University, July 2020 – June 2024 — GPA 8.88/10
- Microsoft Certified: Azure Data Scientist Associate (2026) · AWS Certified Developer – Associate (2026)